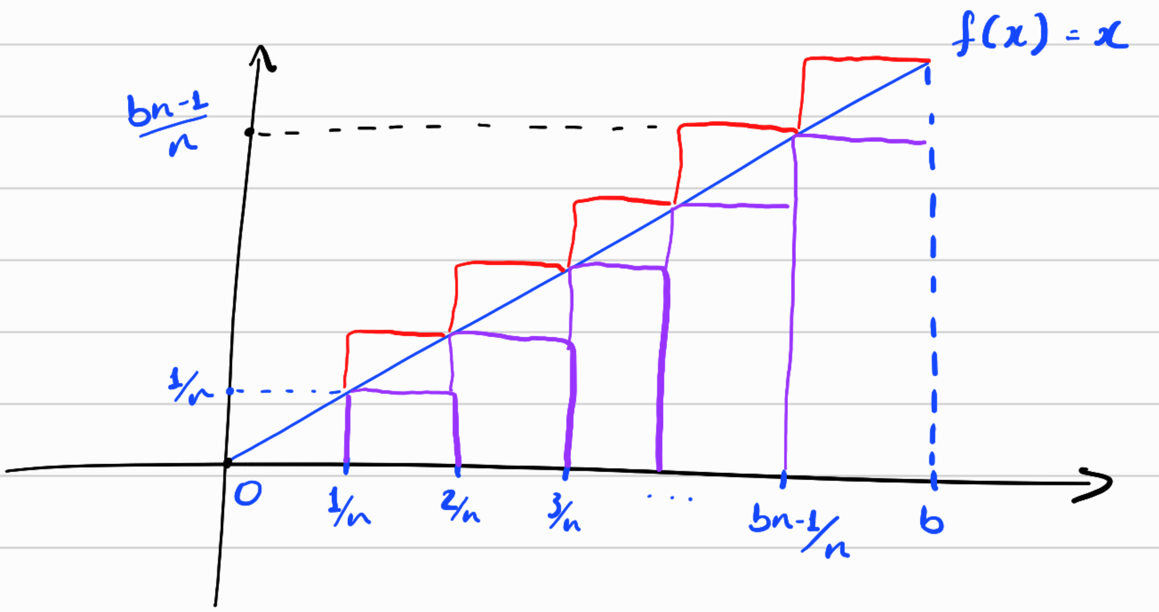
A Integral

$$S_{inf} \leqslant \int_0^b f(x)\,dx \leqslant S_{sup}$$

Integral inferior $\quad \underline{I}(f) = \sup\{S_{inf}\}$

$$\overline{I}(f) = \inf\{S_{sup}\}$$

$\underline{I}(f) = \overline{I}(f)$ se f for integrável



$$S_{inf} = \frac{1}{n}\cdot 0 + \frac{1}{n}\cdot\frac{1}{n} + \frac{1}{n}\cdot\frac{2}{n} + \dots + \frac{1}{n}\cdot\frac{bn-1}{n}$$

$$= \sum_{i=1}^{bn-1}\frac{1}{n}\cdot\frac{i}{n} = \frac{1}{n^2}\sum_{i=1}^{bn-1} i = \frac{1}{n^2}\cdot\frac{(bn-1)\cdot bn}{2} = \frac{b^2}{2} - \frac{b}{2n} \qquad (1)$$

$$S_{sup} = \frac{1}{n}\cdot\frac{1}{n} + \frac{1}{n}\cdot\frac{2}{n} + \dots + \frac{1}{n}\cdot\frac{bn}{n} = \sum_{i=1}^{bn} \frac{1}{n}\cdot\frac{i}{n}$$

$$= \frac{1}{n^2}\sum_{i=1}^{bn} i = \frac{1}{n^2}\cdot\frac{bn(bn+1)}{2} = \frac{b^2}{2} + \frac{b}{2n} \quad (2)$$

Temos também que:

$$S_{inf} \leq \underline{I}(f) \leq \overline{I}(f) \leq S_{sup} \quad (3)$$

É razoável provar isso, pois, por definição, $\begin{cases} \underline{I}(f) = \sup\{S_{inf}\} \\ \overline{I}(f) = \inf\{S_{sup}\} \end{cases}$

Provamos na aula teórica o seguinte resultado:

$$\underbrace{0 \leq \overline{I}(f) - \underline{I}(f) \leq \frac{\overset{\text{constante}}{c}}{n} \quad \forall n}_{*} \quad \Rightarrow \quad \overline{I}(f) = \underline{I}(f)$$

Vamos então mostrar que $*$ é verdade no nosso caso de $f(x) = x$ (generalização ...)

De (3), temos que:

$$\begin{array}{rcccl} -S_{sup} & \leq & -\underline{I}(f) & \leq & -S_{inf} \\ S_{inf} & \leq & \overline{I}(f) & \leq & S_{sup} \\ \hline \end{array}$$

$$S_{inf} - S_{sup} \leq \overline{I}(f) - \underline{I}(f) \leq S_{sup} - S_{inf}$$

de (1) e (2): $S_{inf} - S_{sup} = \frac{b^2}{2} - \frac{b}{2n} - \left(\frac{b^2}{2} + \frac{b}{2n}\right)$

$$-\frac{b}{n} \leq \overline{I}(f) - \underline{I}(f) \leq \frac{b}{n}$$

Portanto,

$$\left| \overline{I}(f) - \underline{I}(f) \right| \leq \frac{b}{n} \quad \forall n.$$

que é justamente * e segue do nosso resultado, portanto, que:

$$\overline{I}(f) = \underline{I}(f) = \int_0^b f(x)\,dx \quad (4)$$

Unindo (1), (2), (3) e (4), temos que:

$$\frac{b^2}{2} - \frac{b}{2n} \leq \int_0^b f(x)\,dx \leq \frac{b^2}{2} + \frac{b}{2n}$$

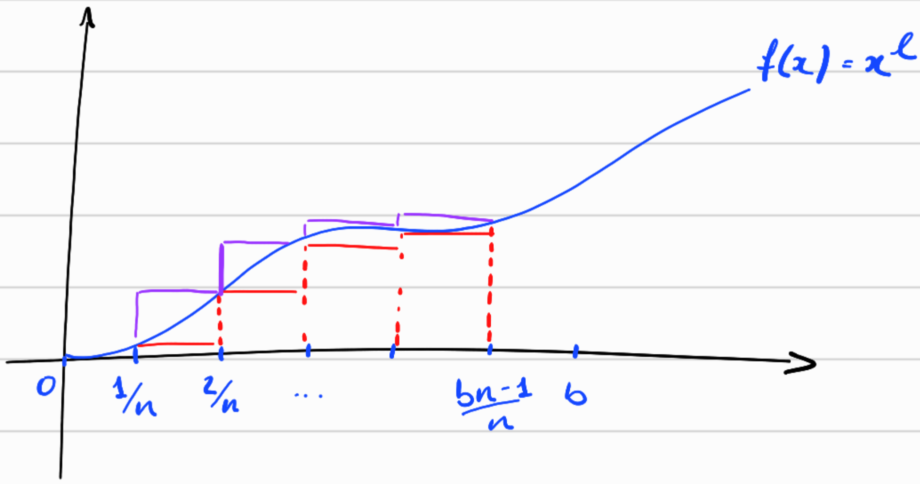
$$-\frac{b}{2n} \leq \int_0^b f(x)\,dx - \frac{b^2}{2} \leq \frac{b}{2n}$$

$$\left| \int_0^b f(x)\,dx - \frac{b^2}{2} \right| \leq \frac{b}{2n}$$

E, pelo lema visto em sala ($\text{se } 0 \leq x \leq c/n \Rightarrow x = 0$), segue que $\boxed{\int_0^b f(x)\,dx = \frac{b^2}{2}}$

Obs. • Nós usamos esse lema p/ concluir que $\overline{I}(f) = \underline{I}(f)$ e agora também p/ concluir o valor da integral.

• Na lista, nós provamos por indução o caso geral



- $$S_{inf} = \cancel{\frac{1}{n}\cdot 0} + \left(\frac{1}{n}\right)^{\ell}\cdot\frac{1}{n} + \left(\frac{2}{n}\right)^{\ell}\cdot\frac{1}{n} + \dots + \left(\frac{bn-1}{n}\right)^{\ell}\cdot\frac{1}{n} +$$

$$= \sum_{i=1}^{bn-1} \frac{1}{n}\cdot\left(\frac{i}{n}\right)^{\ell} = \frac{1}{n^{\ell+1}}\sum_{i=1}^{bn-1} i^{\ell}$$

- $$S_{sup} = \frac{1}{n^{\ell+1}}\sum_{i=1}^{bn} i^{\ell}$$

$$S_{sup} - S_{inf} = \frac{1}{n^{\ell+1}}\sum_{i=1}^{bn} i^{\ell} - \frac{1}{n^{\ell+1}}\sum_{i=1}^{bn-1} i^{\ell} = \cancel{\frac{1}{n^{\ell+1}}\sum_{i=1}^{bn}} i^{\ell} - \left(\cancel{\frac{1}{n^{\ell+1}}\sum_{i=1}^{bn}} i^{\ell} - \frac{b^{\ell}n^{\ell}}{n^{\ell+1}}\right) = \frac{b^{\ell}}{n}$$

Sabemos então que $\bar{I}(f) - \underline{I}(f) \leq b^{\ell}/n$ e, portanto, $\bar{I}(f) = \underline{I}(f)$. Logo,

$$S_{inf} \leq \bar{I}(f) = \underline{I}(f) \leq S_{sup}$$

Para achar o resultado dessa integral, podemos fazer de "dois jeitos":

(1) $$\frac{1}{n^{\ell+1}}\sum_{i=1}^{bn-1} i^{\ell} \leq \bar{I}(f) = \underline{I}(f) \leq \frac{1}{n^{\ell+1}}\sum_{i=1}^{bn} i^{\ell}$$

Pelo resultado $\sum_{i=1}^{n} i^{\ell} = \frac{n^{\ell+1}}{\ell+1} + p_{\ell}(n)$, temos que

$$\frac{1}{n^{\ell+1}}\left(\frac{(bn-1)^{\ell+1}}{\ell+1} + \cancel{p_{\ell}(bn-1)}^{\,0}\right) \leq \bar{I}(f) = \underline{I}(f) \leq \frac{1}{\cancel{n^{\ell+1}}}\left(\frac{b^{\ell+1}\cancel{n^{\ell+1}}}{\ell+1} + \cancel{p_{\ell}(bn)}^{\,0}\right)$$

$$\frac{1}{n^{\ell+1}} \cdot \frac{(bn-1)^{\ell+1}}{\ell+1} \leq \bar{I}(f) = \underline{I}(f) \leq \frac{b^{\ell+1}}{\ell+1}$$

Outra forma seria usando o resultado $\sum_{i=0}^{n-1} i^{\ell} \leq \frac{n^{\ell+1}}{\ell+1} \leq \sum_{i=0}^{n} i^{\ell}$

$$\frac{1}{n^{\ell+1}} \sum_{i=1}^{bn-1} i^{\ell} \leq \underbrace{\bar{I}(f) = \underline{I}(f)}_{I} \leq \frac{1}{n^{\ell+1}} \sum_{i=1}^{bn} i^{\ell}$$

$$\sum_{i=1}^{bn-1} i^{\ell} \leq n^{\ell+1} \cdot I \leq \sum_{i=1}^{bn} i^{\ell} \quad \Rightarrow \quad I = \frac{(bn)^{\ell+1}}{n^{\ell+1} \cdot (\ell+1)} = \frac{b^{\ell+1}}{\ell+1}$$

Portanto, $\bar{I}(f) = \underline{I}(f) = \frac{b^{\ell+1}}{\ell+1}$